IMPLEMENTATION & DEMO

# Course Notes RAG & Adaptive Quiz Engine

A study assistant that answers only from your own lecture PDFs, cites the exact page, and quizzes you on the sections you know least.

BCSE306L Artificial Intelligence · Vellore Institute of Technology

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EVERY ANSWER IS CHECKED

**Verified citation**
Claim found on the cited page

**Auto-cited**
Supported sentence linked to its page

**Not found**
No evidence above τ = 0.35, so no guess

THE PROBLEM

# Generic chatbots don't know your lecture notes

### Answers you can't check

A general chatbot answers from its training data, not the course's notes, and gives no page to verify against.

### Lecture PDFs break naive RAG

Character splitters cut equations in half, scramble table columns, and lose image-only formulas and handwritten scans.

### Revision is passive

Re-reading feels productive but gives no feedback on which sections a student actually knows.

> *Practice testing is one of only two learning techniques rated high utility across ages, materials and tasks.*

Dunlosky et al., Psychological Science in the Public Interest, 2013

# One workspace: library, chat and adaptive quiz

## Library

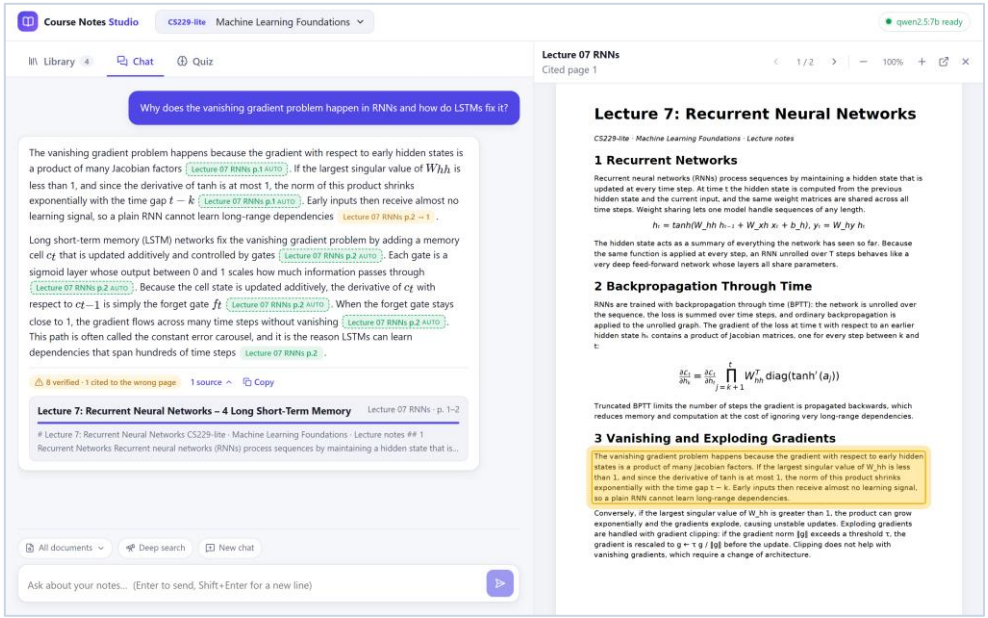
Upload lecture PDFs, Beamer slides or scans. Watch them parse, then preview the extracted Markdown with rendered math.

## Grounded chat

Answers stream in with clickable page citations. A click opens the PDF with the supporting paragraph highlighted.

## Adaptive quiz

Bloom-calibrated multiple-choice questions with a rationale for every option and a mastery score per section.




*Runs fully offline on one laptop: RTX 4070 (8 GB), Qwen 2.5 7B through Ollama.*

HOW IT WORKS

# Five stages from a lecture PDF to a checked answer

### 1 Parse

Layout model finds headings, tables and formulas; OCR reads scans; formulas become LaTeX.

*Docling · RapidOCR*

### 2 Chunk

Sections become parents (≈800–1200 tokens); 200–300-token children with 50-token overlap are indexed.

*Hierarchical chunker*

### 3 Retrieve

Dense and keyword search run in parallel and are fused by rank, then mapped back to parent sections.

*BGE-M3 · bm25s · RRF*

### 4 Rerank

A cross-encoder scores each section against the question; nothing above τ = 0.35 means “Not found”.

*bge-reranker-v2-m3*

### 5 Verify

Qwen writes a cited answer; every citation is then checked against the retrieved pages.

*Qwen 2.5 7B · Ollama*

Python · FastAPI · Qdrant · SQLite · React + Vite · server-sent events for streaming

# Parsing keeps equations, tables and scans intact

### Image-only equation → LaTeX

Bayes' rule recovered as LaTeX from a slide where the formula is only a picture

### Table → Markdown rows

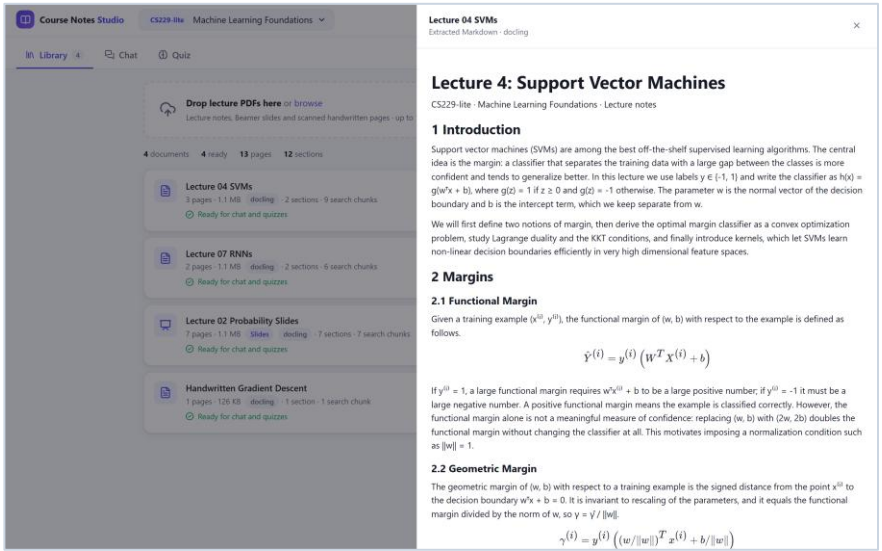
Kernel table keeps its 4 rows × 3 columns; rows are also rewritten as sentences for search

### Handwriting → text

Scanned notes are OCR'd: “If alpha is too large, gradient descent can overshoot…”

### Structure → parent/child chunks

Headings and slides set the parents; a formula or table is never split




**13** pages

**41 s** on the GPU

**23** search chunks

RETRIEVAL

# Hybrid search finds both meaning and exact symbols

**Dense · BGE-M3**
top 25 by meaning

**Keyword · BM25**
top 25; “α” = “\alpha”, “Lemma 3.2”

**Reciprocal Rank Fusion**

$$RRF(d) = \Sigma_m \; 1 \,/\, (60 + rank_m(d))$$

top 15 children → their parent sections, so the model sees whole derivations

**Cross-encoder rerank**
keep top 5 with score > 0.35

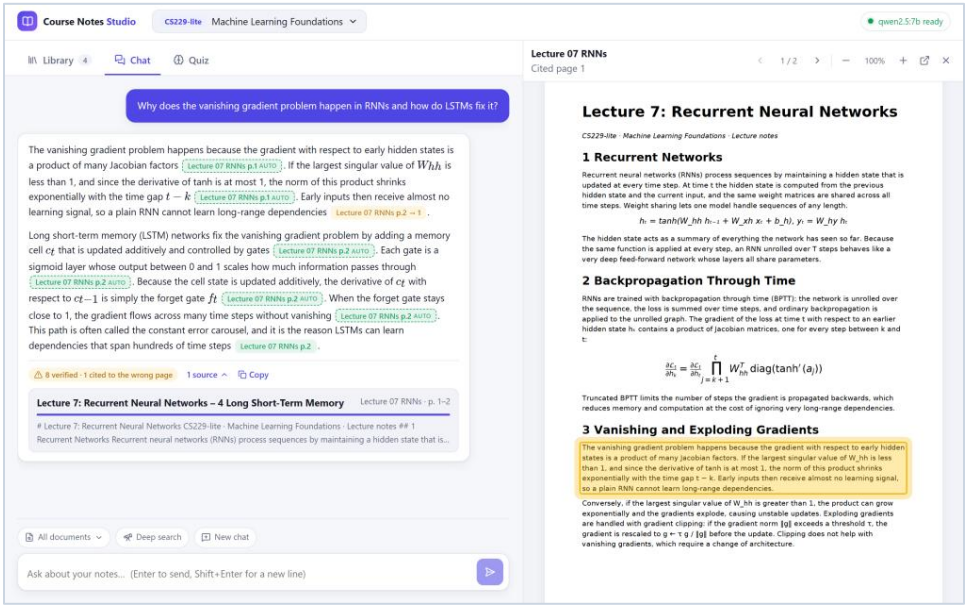
**Nothing passes?**
reply “Not found in course notes”

## Why both?

Embeddings understand paraphrases (“why do early inputs stop learning?”) but blur exact tokens; BM25 with a math-aware tokenizer catches theorem numbers, variable names and Greek letters. Fusing ranks, not scores, needs no tuning between the two.

GROUNDED CHAT

# Every citation is checked against the retrieved pages




1 **Source**
the cited file and page must be in the retrieved context

2 **Page**
if the claim's words are on another page, the chip shows p.2 → 1

3 **Highlight**
the viewer opens on the single paragraph that matches the claim

4 **Completion**
uncited but supported sentences get an auto citation; the rest are flagged as unsupported

**Sentences with a verified citation (Qwen 2.5 7B): 29% → 100%**

# Every question is repaired and validated first

### Pick sections

70% weighted toward weak sections, 30% by relevance

### Number the evidence

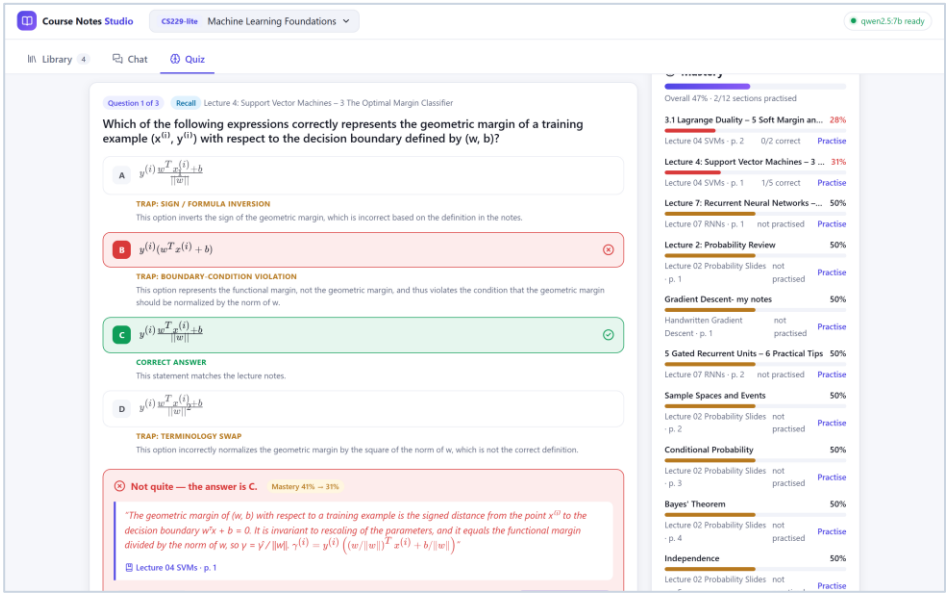
notes shown as [S1], [S2]… so Qwen cites sentence ids, not a retyped quote

### Generate to a schema

4 options; each distractor labelled sign inversion, boundary violation or terminology swap

### Repair and validate

fix label slips; reject if not exactly one correct, bad LaTeX, or unsupported evidence




*Questions passing validation: 33% with free-text quotes → 67% with evidence ids → 100% (9 of 9) with repair*

ADAPTIVE PRACTICE

# Weak sections come back far more often

**After each first attempt**

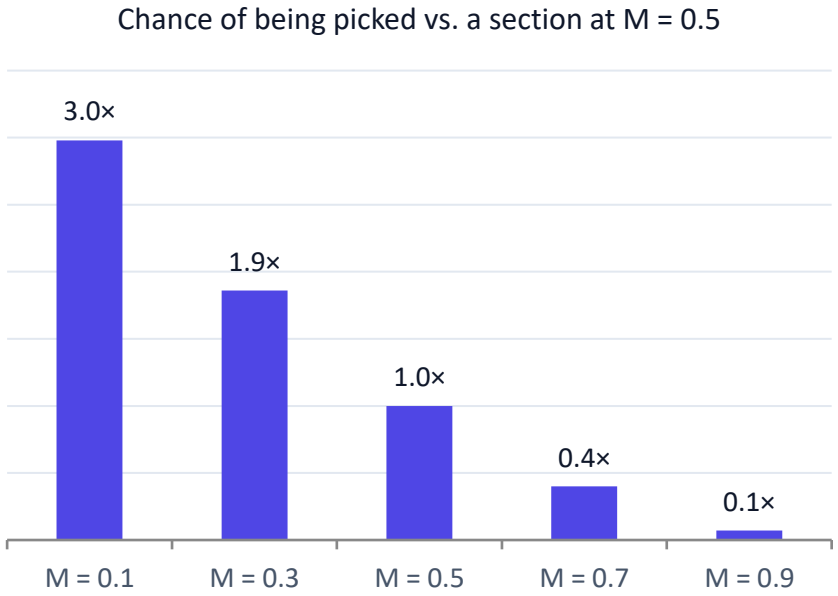
$$M \leftarrow clip(M + \alpha(S - M), 0, 1)$$

S = 1 if correct, 0 if not · α = 0.25 · M starts at 0.5

**Choosing the next quiz's sections**

$$P(\text{section } c) \propto (1.05 - M_c)^2$$

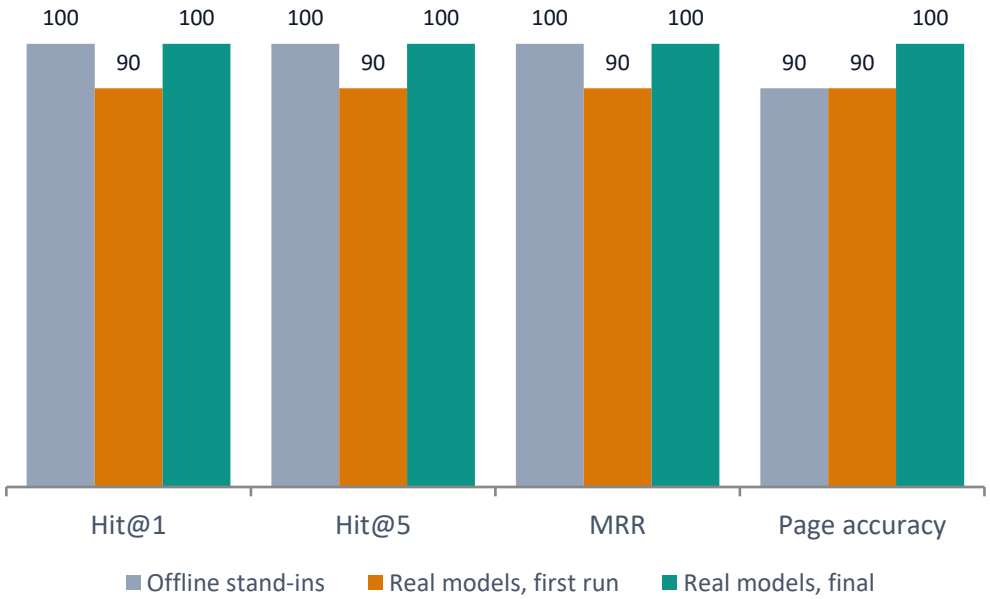
for 70% of slots; the other 30% keep coverage




*A section at 10% mastery is picked about 40× more often than one at 90%.*

RESULTS

# Right lecture section found for 10 of 10 questions




| Measure | Final |
|---|---|
| Hit@1 / Hit@5 | 100% / 100% |
| Mean reciprocal rank | 1.00 |
| Cited page inside section | 100% |
| Warm query latency | 268 ms |

The one miss in the first run was “variance of a Bernoulli variable”, answered by a table. Rewriting table rows as sentences for embedding and reranking fixed it (score 0.38 > τ), and lifted the kernel-table question from 0.61 to 0.86.

*10 hand-written questions over 4 lecture PDFs (13 pages): paraphrases, theorem numbers, a table lookup and a handwritten scan.*

RESULTS

# What made a 7B model reliable enough

**Sentences with a verified citation**

rule repeated in the user turn + server-side completion

29% → **100%**

**Quiz questions passing validation**

numbered evidence sentences + repair of label slips

33% → **100%**

**Benchmark questions found (Hit@1)**

table rows rewritten as sentences for search

90% → **100%**

AUTOMATED TESTS, ALL PASSING

| | |
|---|---|
| **289** | backend unit + API end-to-end |
| **56** | frontend unit (Vitest) |
| **9** | browser end-to-end (desktop + mobile) |
| **5** | live tests on the real models |

*Positive and negative cases for every component: invalid, empty, encrypted and duplicate PDFs, off-topic questions, malformed model output.*

RELATION TO THE SYLLABUS

# Modules 1–4: agents, search, local search, logic

| Module | Syllabus topics | Where it appears in this project | Link |
|---|---|---|---|
| **M1 Introduction & Intelligent Agents** | Agent structure, environments, applications of AI | The app is a learning agent: it perceives notes (parser, OCR) and questions, acts by answering and quizzing, and keeps a student model. The environment is partially observable (knowledge is hidden) and sequential. | **Strong** |
| **M2 Search** | State-space search, best-first, uniform cost, A* | Retrieval is search over a space of passages: Qdrant's HNSW index runs greedy best-first search on vector distance; BM25 and RRF rank and prune candidates to the top 15 / top 5. | **Strong** |
| **M3 Local & Adversarial Search** | Hill climbing, simulated annealing, minimax | Quiz generation is generate-and-test: score a draft against constraints, repair it, or restart with a higher temperature. Distractors are chosen as the most tempting wrong answers. | **Partial** |
| **M4 Logic & Reasoning** | Propositional and first-order logic, forward chaining | Citation and quiz validation are rule-based inference: facts (file, page, section range) fire rules that conclude valid / moved / unsupported; “exactly one correct option” is a propositional constraint. | **Strong** |

# Modules 5–8: uncertainty, planning, language

| Module | Syllabus topics | Where it appears in this project | Link |
|---|---|---|---|
| **M5 Uncertain Knowledge** | Quantifying uncertainty, Bayes rule, Bayesian networks | BM25 comes from the probabilistic relevance model; the reranker outputs P(relevant \| question, passage) and τ = 0.35 is a decision threshold; mastery is a belief about the student, updated by evidence (a simplified knowledge-tracing model). | **Strong** |
| **M6 Planning** | Classical, hierarchical planning, acting under uncertainty | A quiz is planned top-down: goal (n questions on weak spots) → slots → section → Bloom level → generate → validate. Ingestion is a tracked plan (queued → parsing → chunking → indexing → ready) with a fallback parser. | **Partial** |
| **M7 Communicating, Perceiving, Acting** | Probabilistic language processing, information retrieval, information extraction, perception, object recognition | The core of the project: hybrid retrieval and reranking (IR), layout parsing, tables and formula-to-LaTeX (IE), LLM generation and tokenization (language), layout detection and OCR on page images (perception). | **Strong** |
| **M8 Contemporary Issues** | Current topics in AI | Hallucination and grounded generation, verifiable citations, local and private inference on a laptop GPU, academic integrity. | **Strong** |

RELATION TO THE SYLLABUS

# Course outcomes met by the project

**CO1** **Evaluate AI methods and their foundations**

10-question retrieval benchmark (Hit@k, MRR); offline stand-ins vs. real models; before/after ablations of each fix.

**CO2** **Apply AI to inference, perception, representation and learning**

Perception (layout, OCR), knowledge representation (parent/child chunks, vectors), inference (verification rules), learning (mastery model).

**CO3** **Reason with uncertainty and knowledge representation**

Probabilistic ranking, a relevance threshold that answers “Not found” instead of guessing, mastery as an updated belief.

**CO4** **Show how search drives problem solving**

Best-first vector search, BM25 ranking, rank fusion and top-k pruning decide what the model is allowed to say.

WHAT WE LEARNED

# Problems we hit, and how we fixed them

### Quotes were paraphrased

Qwen rewrote the “exact quote”, so 2 of 3 questions failed. Showing the notes as numbered sentences and asking for ids makes quotes verbatim by construction.

### One citation per paragraph

Only 29% of sentences carried a citation. Repeating the rule in the user turn and completing citations on the server reached 100%.

### Tables were invisible to search

Pipes and dashes don't look like a question. Rewriting rows as “Distribution: Bernoulli; Variance: p(1 − p).” fixed the benchmark miss.

### Every heading was level 1

Docling flattens heading levels, so sections never merged into 800-token parents. Merging now stops only at “Lecture N” / “Chapter N”.

LIMITATIONS & NEXT STEPS

# What it can't do yet, and where it goes next

### Bigger evaluation

A larger, independently written question set per course, and quiz-quality ratings from students and faculty.

### Stronger verification

Use an entailment (NLI) model next to the lexical check, so paraphrased claims are verified too.

### Better student model

Replace the moving-average mastery with Bayesian knowledge tracing and spaced-repetition scheduling.

## KNOWN LIMITATIONS

- Evaluation scale: 10 benchmark questions over 4 synthetic lecture PDFs; results need confirming on real course material.
- A 7B model still varies run to run: answers may cite few sentences (completion fills the gap), and two distractors can look nearly identical once rendered.
- Formula OCR can misread symbols (ŷ read as Ŷ), and the lexical page check cannot verify paraphrased claims.

THANK YOU

# Questions and feedback welcome

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github.com/RoyceRRNS/AI_Project

